

The total x -current density is $\mathbf{j}_{x2}$, where

$$\begin{aligned}\mathbf{j}_{x1} &= (\mathbf{n} \times \mathbf{j}_{ey1})/\omega_e \tau_{en} - (\mathbf{n} \times \mathbf{j}_{iy1})/\omega_i \tau_{in} \\ &= (\sigma_{en}/\omega_e^3 \tau_{en}^3 - \sigma_{in}/\omega_i^3 \tau_{in}^3) \mathbf{n} \times \mathbf{E}.\end{aligned}\quad (29)$$

It should be noted that resistance due to electron-ion collisions makes *no* contribution to the x current because the drifts due to equal and opposite forces on electrons and ions are always equal.

Resistance to the x current will set up a further y current, $\mathbf{j}_{ey3}$ for electrons and $\mathbf{j}_{iy3}$ for ions. In magnitude, these currents are given by

$$\begin{aligned}\mathbf{j}_{ey3} &= (\omega_e \tau_{en})^{-1} \mathbf{n} \times \mathbf{j}_{ex2} \\ &= -(\omega_e \tau_{en})^{-2} \mathbf{j}_{ey1}, \\ \mathbf{j}_{iy3} &= (\omega_i \tau_{in})^{-1} \mathbf{n} \times \mathbf{j}_{ix2} \\ &= -(\omega_i \tau_{in})^{-2} \mathbf{j}_{iy1}.\end{aligned}\quad (30)$$

The sequence of reactions continues to infinity. To obtain the total current density in the y direction due to electrons, we must sum an infinite geometric series with common ratio $-(\omega_e \tau_{en})^{-2}$. The sum is the first term multiplied by $(1 + \omega_e^{-2} \tau_{en}^{-2})^{-1}$. A similar series must be summed to obtain the total y -current density due to ions. Again, two further such series must be summed for the current density in the x direction. The

final result is

$$\begin{aligned}\mathbf{j}_y &= \{[\sigma_{en}/(1 + \omega_e^2 \tau_{en}^2)] + [\sigma_i/(1 + \omega_i^2 \tau_{in}^2)]\} \mathbf{E}, \\ \mathbf{j}_x &= \{[\sigma_{en}/\omega_e \tau_{en}(1 + \omega_e^2 \tau_{en}^2)] \\ &\quad - [\sigma_{in}/\omega_i \tau_{in}(1 + \omega_i^2 \tau_{in}^2)]\} \mathbf{n} \times \mathbf{E}.\end{aligned}\quad (31)$$

The geometrical series fail to converge when $(\omega_e \tau_{en})^{-1} > 1$ and $(\omega_i \tau_{in})^{-1} > 1$. The reason is that the drift momenta $m\mathbf{V}_E$, $m\mathbf{V}_{y1}$, $m\mathbf{V}_{x2}$, $m\mathbf{V}_{y3}$, etc. are acquired only after a period of order ω^{-1} following an impulse. These drift momenta cannot, therefore, be lost more rapidly than once per interval ω^{-1} . When τ falls below this period, reanalysis of the situation is required.

The case $\omega\tau < 1$ has been treated, from the point of view of orbit theory, by Chapman and Cowling.² It is remarkable, in view of their quite different approach, that the result should be identical to Eqs. (31)—the second of Eqs. (31) becomes Chapman and Cowling's equation for the current in the direction of the drift on noting that $\sigma_{en}(\omega_e \tau_{en})^{-1} = \sigma_{in}(\omega_i \tau_{in})^{-1}$.

¹ L. Spitzer, *Physics of Fully Ionized Gases*, (Wiley-Interscience, New York, 1956), pp. 18 and 26.

² S. Chapman and T. G. Cowling, *The Mathematical Theory of Non-Uniform Gases*, (Cambridge U. P., Cambridge, Eng., 1952), pp. 325-29.

A Derivation of the Macroscopic Maxwell Equations*

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We derive the macroscopic Maxwell equations from the microscopic equations of electrodynamics. The method is based on a modified version of spatial averaging and unifies several past approaches. For simplicity, the derivation presented here is restricted to classical systems, although it may easily be extended to incorporate quantum mechanical systems.

I. INTRODUCTION

To an excellent approximation matter may be regarded as being composed of point electrons, point nuclei, and microscopic electric and magnetic fields. The microscopic equations of electro-

dynamics predict in detail the behavior of the particles and their fields. The *macroscopic* equations of electrodynamics (the macroscopic Maxwell equations) predict the *average* behavior of the particles and the fields. The macroscopic equations may be derived from the microscopic

equations by a suitable averaging procedure, a view developed initially by Lorentz. Subsequently, there have been many attempts to provide derivations of increasing rigor, such as those found in the works of Van Vleck,¹ Rosenfeld,² Mazur and Nijboer,³ and De Groot.⁴

Yet another approach will be presented here, whose fundamental assumption is only that macroscopic quantities are spatial averages of the corresponding microscopic quantities. Past analyses have incorporated one or more combinations of three types of averaging procedures, viz., spatial averaging, time averaging, and ensemble averaging. It is clearly useful to obtain macroscopic quantities from their microscopic counterparts with the use of a minimum number of averaging procedures since such a method will have the widest applicability. We will now argue that of the three we must necessarily retain spatial averaging.

Consider the microscopic electric field inside a solid (an ionic crystal is a good example). On an atomic scale, the microscopic electric field is obviously a rapidly fluctuating function of position. If we examine the field at a fixed point, we find that it will vary in time because the particles in the solid vibrate relative to their average positions. The *time average* of the field at the point is obtained by integrating the time varying field at this point over some time interval and dividing by the time span. The *ensemble average* of the field at a point is obtained by calculating the field at this point due to some configuration (positions and momenta) of the particles, and then averaging over the possible configurations according to some probability distribution (Boltzmann's law, say). Because in solids the particles do not move much relative to their average positions, the time or ensemble averaged electric field is *still* a rapidly fluctuating function of position and as such, is not an acceptable macroscopic field. The use of spatial averaging to smooth out these fluctuations is therefore unavoidable.

It is often stated that macroscopic quantities are averages over volumes large compared to atomic volumes and over times long compared to atomic times. The inclusion of time averaging along with spatial averaging is not always correct. For example, optical phenomena, which are frequently investigated with the use of the macro-

scopic Maxwell equations, often have the same frequency as atomic motions. If time averaging were performed, all of the quantities of interest would be averaged to zero, a physically uninteresting result. In order to keep our derivation as general as possible, we will therefore exclude time averaging. Furthermore, in the interest of generality, we will not ensemble average. At the completion of our derivation, we will return to the question of possible additional averages.

We will show that the conventional definition of spatial average (essentially that advocated by Lorentz and by Rosenfeld) is defective, and we will substitute for it a modified, more rigorous version. The formal appearance of the resulting equations will resemble those of Mazur and Nijboer and De Groot. The results presented here may, in a sense, be regarded as providing a unification of past approaches which are seemingly different.

In the interests of presenting a proof which is as straightforward as possible, we will neglect the complications of quantum mechanics and will confine our analysis to classical systems. However, the mathematical formalism and the method of analysis is such that, with minor changes, a quantum mechanical derivation can easily be constructed. The results of the quantum mechanical derivation are essentially identical to the classical results and will not be given here.⁵

II. DISCUSSION OF SPATIAL AVERAGES

The conventional definition of a spatial average of a microscopic function of position $A(\mathbf{x})$ is

$$\langle A(\mathbf{x}) \rangle = \int d^3s w(\mathbf{s}) A(\mathbf{x} - \mathbf{s}), \quad (1)$$

where $w(\mathbf{s})$ is a real positive function which is constant in some volume of space surrounding $\mathbf{s} = 0$ and is zero otherwise. It is normalized to unity,

$$\int d^3s w(\mathbf{s}) = 1$$

and has dimensions large compared to molecular sizes. As an example,

$$w(\mathbf{s}) = (\frac{4}{3}\pi R^3)^{-1} \quad \text{if } |\mathbf{s}| < R$$

and

$$w(\mathbf{s}) = 0 \quad \text{if } |\mathbf{s}| > R.$$

Although not specifically written down in the manner of Eq. (1), this definition of spatial aver-

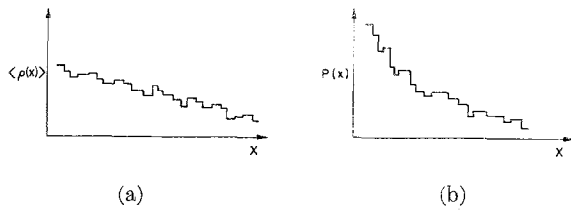


FIG. 1. The average charge density as a function of position (a) and the average polarization as a function of position (b). The figures are for a one dimensional system.

age is essentially that used by Lorentz and Rosenfeld.² However, the form for the weight function $w(\mathbf{s})$ is quite often inadequate (as we shall see) and must be modified.

Consider, for example, a system composed of neutral dipoles spatially distributed in any manner. The average charge density and average polarization as a function of position [and as defined by Eq. (1)] are shown schematically in Fig. 1.

These averages are clearly not smooth but have a steplike form for the following reason: Whenever a charge (or a dipole) enters the volume [defined by $\omega(\mathbf{x}-\mathbf{s}) \neq 0$], the average charge density (or polarization) changes by a discrete and finite amount at $\mathbf{x}$. The interpretation of the equation $\langle \rho \rangle \approx -\nabla \cdot \mathbf{P}$ is obscured by the fact that $(-\nabla \cdot \mathbf{P})$ is a wildly fluctuating function of position and is thus a poor representative of the average density. The large fluctuations in $\nabla \cdot \mathbf{P}$ are due of course to the steplike behavior of $\mathbf{P}$. It is quite well understood in physical terms that what is required is a mechanism to smooth the charge density and the polarization [as shown in Fig. (2)], so that the equation $\langle \rho \rangle \approx -\nabla \cdot \mathbf{P}$ becomes physically meaningful and valid. Time averaging or ensemble averaging are often invoked to accomplish the necessary smoothing.

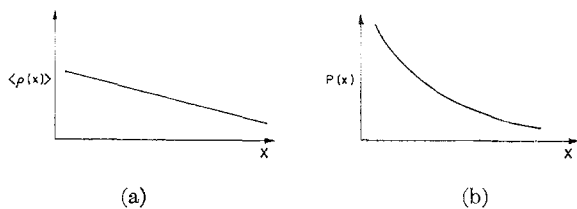


FIG. 2. The smoothed average charge density as a function of position (a) and the smoothed average polarization as a function of position (b). The figures are for a one dimensional system.

In order to eradicate the steplike behavior, we propose instead to modify the definition of $w(\mathbf{s})$ yet leave the physical meaning of spatial average unaltered. Instead of permitting $w(\mathbf{s})$ to go abruptly to zero at the boundary of the averaging volume, we shall require the function to monotonically approach zero in some well-behaved fashion and at the same time demand that the transition region have a size large compared to molecular dimensions. A comparison of the old and new forms are shown in Fig. 3.

With this minor modification, the averages are now smooth functions of position as in Fig. 2 since charges no longer abruptly enter or leave a sharply defined region. We also note that the averages are also smooth functions of time. If, for example, a charge were to enter a sharply defined region of space, $\langle \rho \rangle$ would change in time in a

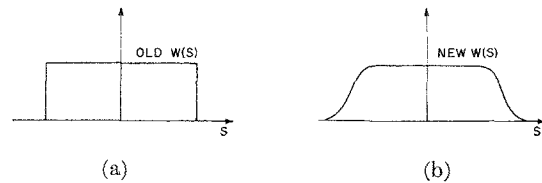


FIG. 3. Conventional definition of $w(s)$ (a) and modified definition of $w(s)$ (b). The figures are for a one dimensional system.

steplike manner, and $(d/dt) \langle \rho \rangle$ would therefore be a sharply fluctuating function of time. However, with our modified form of $w(\mathbf{s})$, $(d/dt) \langle \rho \rangle$ is now a smooth, well-behaved function of time.

Now that we have demonstrated the necessity of modifying the definition of the spatial average and have indicated the general form of this modification, we will turn to an explicit definition of spatial average.

III. DEFINITION OF SPATIAL AVERAGE

The spatial average of a microscopic function of position $A(\mathbf{x})$ is defined as

$$\langle A(\mathbf{x}) \rangle = \int d^3s w(\mathbf{s}) A(\mathbf{x}-\mathbf{s}), \quad (2)$$

where $w(\mathbf{s})$ is a real, positive, well-behaved function which (according to the previous discussion) is nonzero only in some region of space surrounding $\mathbf{s}=0$. It is normalized to unity,

$$\int d^3s w(\mathbf{s}) = 1$$

TABLE I. List of symbols and definitions.

Symbol	Definition	Description
$\mathbf{x}_j$		Position of the j th particle
$\mathbf{x}_l$		Center of mass position of the l th molecule (or other convenient point within the molecule)
$\mathbf{v}_j$	$\langle d/dt \rangle \mathbf{x}_j$	Velocity of the j th particle
$\mathbf{v}_l$	$\langle d/dt \rangle \mathbf{x}_l$	Center of mass velocity of the l th molecule
$\mathbf{x}_{jl}$	$\mathbf{x}_j - \mathbf{x}_l$	
$\mathbf{v}_{jl}$	$\mathbf{v}_j - \mathbf{v}_l$	
$\sum_{j(l)}$		Sum over the constituent particles the l th molecule
q_j		Charge of the j th particle
q_l	$\sum_{j(l)} q_j$	Charge of the l th molecule
$\mathbf{p}_l$	$\sum_{j(l)} q_j \mathbf{x}_{jl}$	Dipole moment of the l th molecule
$\mathbf{q}_l$	$\frac{1}{2} \sum_{j(l)} q_j \mathbf{x}_{jl} \mathbf{x}_{jl}$	Quadrupole moment of the l th molecule
$\mathbf{m}_l$	$(2c)^{-1} \sum_{j(l)} q_j \mathbf{x}_{jl} \times \mathbf{v}_{jl}$	Magnetic moment of the l th molecule
$\rho(\mathbf{x})$	$\sum_j q_j \delta(\mathbf{x} - \mathbf{x}_j)$	Microscopic charge density
$\mathbf{j}(\mathbf{x})$	$\sum_j q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{x}_j)$	Microscopic current density
$\sum_{(ee)}$		Sum over the conduction electrons
$\langle \quad \rangle$		Spatial average
$\rho_F(\mathbf{x})$	$\langle \sum_{j(ee)} q_j \delta(\mathbf{x} - \mathbf{x}_j) \rangle + \langle \sum_l q_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle$	"Free" charge density
$\mathbf{P}(\mathbf{x})$	$\langle \sum_l \mathbf{p}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle$	Polarization
$\mathbf{Q}(\mathbf{x})$	$\langle \sum_l \mathbf{q}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle$	Quadrupole moment density
$\mathbf{J}_F(\mathbf{x})$	$\langle \sum_{j(ee)} q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{x}_j) \rangle + \langle \sum_l q_l \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle$	"Free" current density
$\mathbf{M}(\mathbf{x})$	$\langle \sum_l \mathbf{m}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle$	Magnetization
$\mathbf{e}$		Microscopic electric field
$\mathbf{b}$		Microscopic magnetic field
$\mathbf{E}$	$\langle \mathbf{e} \rangle$	Macroscopic electric field
$\mathbf{B}$	$\langle \mathbf{b} \rangle$	Macroscopic magnetic field

and varies sufficiently slowly that the approximation

$$w(\mathbf{s}+1) \approx w(\mathbf{s}) + (\mathbf{1} \cdot \nabla_{\mathbf{s}})w(\mathbf{s}) + \frac{1}{2}(\mathbf{1} \cdot \nabla_{\mathbf{s}})^2 w(\mathbf{s}) \quad (3)$$

is well obeyed for $\mathbf{1}$ of the order of molecular sizes. We will see that Eq. (3) will allow us to make the necessary multipole moment expansions.

The state of a system is specified by the positions and velocities of the electrons and nuclei and by the values of the microscopic electric and

magnetic fields. It is assumed that some of the particles can be grouped together (or localized) into small stable entities (which we call molecules) while the other particles (which we call conduction electrons) cannot be. This division of charges into "bound" charge and "unbound" charge is of course arbitrary to some extent.⁶ However, it is well known that for many physical systems this division can be achieved, and we will assume that this is the case here.

The microscopic equations of electrodynamic

are:

$$\nabla \times \mathbf{e} + c^{-1}(d/dt)\mathbf{b} = 0, \quad (4a)$$

$$\nabla \cdot \mathbf{b} = 0, \quad (4b)$$

$$\nabla \cdot \mathbf{e} = 4\pi\rho, \quad (4c)$$

$$\nabla \times \mathbf{b} = (4\pi/c)\mathbf{j} + c^{-1}(d/dt)\mathbf{e}, \quad (4d)$$

where $\mathbf{e}$, $\mathbf{b}$, ρ , and $\mathbf{j}$ are, respectively, the microscopic electric field, magnetic field, charge density, and current density (see Table I for a list of definitions).

It is straightforward to show from Eq. (2) that differentiation and averaging may be interchanged:

$$(d/dt)\langle A(\mathbf{x}) \rangle = \langle (dA/dt)(\mathbf{x}) \rangle, \quad (5a)$$

$$(d/dx_i)\langle A(\mathbf{x}) \rangle = \langle (dA/dx_i)(\mathbf{x}) \rangle. \quad (5b)$$

We will denote the macroscopic electric field and the macroscopic magnetic field by $\mathbf{E} = \langle \mathbf{e} \rangle$ and $\mathbf{B} = \langle \mathbf{b} \rangle$. If we average Eq. (4) and use Eq. (5), we obtain the averaged equations of electrodynamics:

$$\nabla \times \mathbf{E} + c^{-1}(d/dt)\mathbf{B} = 0, \quad (6a)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (6b)$$

$$\nabla \cdot \mathbf{E} = 4\pi\langle \rho \rangle, \quad (6c)$$

$$\nabla \times \mathbf{B} = (4\pi/c)\langle \mathbf{j} \rangle + c^{-1}(d/dt)\mathbf{E}. \quad (6d)$$

The purpose of averaging the microscopic equations is to obtain approximate (but potentially useful) expressions for $\langle \rho \rangle$ and $\langle \mathbf{j} \rangle$ in terms of quantities which depend on internal molecular moments. In the calculations which follow, the approximations will include effects up to and including electric quadrupole moments and magnetic dipole moments, but nothing of a higher order. These approximations will be more than sufficient for most applications.

IV. AVERAGE CHARGE DENSITY

As we have remarked, the charge density can be split into those charges which belong to the conduction (unbound) electrons and those (bound) charges which belong to the molecules. The latter may be further classified according to which mole-

cule the charge belongs. We may therefore write

$$\begin{aligned} \rho(\mathbf{x}) &= \sum_j q_j \delta(\mathbf{x} - \mathbf{x}_j) \\ &= \sum_{j(\text{oe})} q_j \delta(\mathbf{x} - \mathbf{x}_j) + \sum_{j(\text{mol})} q_j \delta(\mathbf{x} - \mathbf{x}_j) \\ &= \sum_{j(\text{oe})} q_j \delta(\mathbf{x} - \mathbf{x}_j) + \sum_l \sum_{j(l)} q_j \delta(\mathbf{x} - \mathbf{x}_j). \end{aligned} \quad (7)$$

Consider first the average charge density of the l th molecule. From Eqs. (2) and (7),

$$\begin{aligned} \langle \rho_l(\mathbf{x}) \rangle &= \int d^3s w(\mathbf{s}) \rho_l(\mathbf{x} - \mathbf{s}) \\ &= \int d^3s w(\mathbf{s}) \sum_{j(l)} q_j \delta(\mathbf{x} - \mathbf{s} - \mathbf{x}_j) \\ &= \sum_{j(l)} q_j w(\mathbf{x} - \mathbf{x}_l - \mathbf{x}_{jl}). \end{aligned} \quad (8)$$

Since $\mathbf{x}_{jl} = \mathbf{x}_j - \mathbf{x}_l$ is the distance between a charge in a molecule and the center of mass of the same molecule, it follows that $\mathbf{x}_{jl}$ is of molecular size. A Taylor expansion of w may now be made about the point $\mathbf{x} - \mathbf{x}_l$ [see Eq. (3)] in accordance with the discussion following Eq. (2). (It is at this point that the present derivation makes contact with the work of Mazur and Nijboer and De Groot). If as usual we denote $d/d\mathbf{x}$ by ∇ , we have

$$\begin{aligned} \langle \rho_l(\mathbf{x}) \rangle &\approx \sum_{j(l)} q_j [w(\mathbf{x} - \mathbf{x}_l) - (\mathbf{x}_{jl} \cdot \nabla) w(\mathbf{x} - \mathbf{x}_l) \\ &\quad + \frac{1}{2} (\mathbf{x}_{jl} \cdot \nabla)^2 w(\mathbf{x} - \mathbf{x}_l)] \\ &= q_l w(\mathbf{x} - \mathbf{x}_l) - \mathbf{p}_l \cdot \nabla w(\mathbf{x} - \mathbf{x}_l) \\ &\quad + \mathbf{q}_l \cdot \nabla \cdot \nabla w(\mathbf{x} - \mathbf{x}_l) \\ &= \int d^3s w(\mathbf{s}) q_l \delta(\mathbf{x} - \mathbf{s} - \mathbf{x}_l) \\ &\quad - \nabla \cdot \int d^3s w(\mathbf{s}) \mathbf{p}_l \delta(\mathbf{x} - \mathbf{s} - \mathbf{x}_l) \\ &\quad + \nabla \cdot \nabla \cdot \int d^3s w(\mathbf{s}) \mathbf{q}_l \delta(\mathbf{x} - \mathbf{s} - \mathbf{x}_l) \\ &= \langle q_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle - \nabla \cdot \langle \mathbf{p}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad + \nabla \cdot \nabla \cdot \langle \mathbf{q}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle. \end{aligned} \quad (9)$$

The average charge density is obtained by combining Eqs. (7) and (9):

$$\begin{aligned} \langle \rho(\mathbf{x}) \rangle &= \langle \sum_{j(\text{oe})} q_j \delta(\mathbf{x} - \mathbf{x}_j) \rangle + \langle \sum_l \rho_l(\mathbf{x}) \rangle \\ &= \langle \sum_{j(\text{oe})} q_j \delta(\mathbf{x} - \mathbf{x}_j) \rangle + \langle \sum_l q_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad - \nabla \cdot \langle \sum_l \mathbf{p}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad + \nabla \cdot \nabla \cdot \langle \sum_l \mathbf{q}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &= \rho_F(\mathbf{x}) - \nabla \cdot [\mathbf{P}(\mathbf{x}) - \nabla \cdot \mathbf{Q}(\mathbf{x})]. \end{aligned} \quad (10)$$

This result is what we have set out to calculate: The average charge density is expressed in terms of quantities which depend on the internal moments of the molecules. We will now apply the same averaging procedure to the current density.

V. AVERAGE CURRENT DENSITY

The current density, which is given by

$$\mathbf{j}(\mathbf{x}) = \sum_j q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{x}_j); \quad [\mathbf{v}_j = (d/dt)\mathbf{x}_j],$$

can be split into parts whose charges belong to the conduction electrons and whose charges belong to the molecules [as in Eq. (7)]. We will first analyze the current density of the l th molecule⁷:

$$\begin{aligned} \langle \mathbf{j}_l(\mathbf{x}) \rangle &= \int d^3s w(\mathbf{s}) \mathbf{j}_l(\mathbf{x} - \mathbf{s}) \\ &= \int d^3s w(\mathbf{s}) \sum_{j(l)} q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{s} - \mathbf{x}_j) \\ &= \sum_{j(l)} q_j (\mathbf{v}_{jl} + \mathbf{v}_l) w(\mathbf{x} - \mathbf{x}_l - \mathbf{x}_{jl}). \end{aligned}$$

Again, a Taylor series expansion of w is now made about the point $\mathbf{x} - \mathbf{x}_l$, with the result

$$\begin{aligned} \langle \mathbf{j}_l(\mathbf{x}) \rangle &\approx q_l \mathbf{v}_l w(\mathbf{x} - \mathbf{x}_l) + [(d/dt)\mathbf{p}_l] w(\mathbf{x} - \mathbf{x}_l) \\ &\quad - \mathbf{v}_l (\mathbf{p}_l \cdot \nabla) w(\mathbf{x} - \mathbf{x}_l) \\ &\quad + \mathbf{v}_l (\mathbf{q}_l \cdot \nabla \cdot \nabla) w(\mathbf{x} - \mathbf{x}_l) \\ &\quad - \sum_{j(l)} q_j \mathbf{v}_{jl} (\mathbf{x}_{jl} \cdot \nabla) w(\mathbf{x} - \mathbf{x}_l). \end{aligned} \quad (11)$$

A third-order molecular moment term has been dropped from Eq. (11). The following two equations may be directly verified⁸:

$$\begin{aligned} (d/dt)(\mathbf{p}_l w) + \nabla \times (\mathbf{p}_l \times \mathbf{v}_l w) \\ &= [(d/dt)\mathbf{p}_l] w - \mathbf{v}_l (\mathbf{p}_l \cdot \nabla) w, \quad (12a) \\ \nabla \times (c\mathbf{m}_l w) - (d/dt)(\mathbf{q}_l \cdot \nabla w) - \nabla \times \nabla \cdot (\mathbf{q}_l \times \mathbf{v}_l w) \\ &= \mathbf{v}_l (\mathbf{q}_l \cdot \nabla \cdot \nabla) w - \sum_{j(l)} q_j \mathbf{v}_{jl} (\mathbf{x}_{jl} \cdot \nabla) w. \end{aligned} \quad (12b)$$

If we substitute Eqs. (12) into Eq. (11) we obtain the average current density of the l th mole-

cule:

$$\begin{aligned} \langle \mathbf{j}_l(\mathbf{x}) \rangle &= q_l \mathbf{v}_l w(\mathbf{x} - \mathbf{x}_l) + (d/dt)\mathbf{p}_l w(\mathbf{x} - \mathbf{x}_l) \\ &\quad - (d/dt)\mathbf{q}_l \cdot \nabla w(\mathbf{x} - \mathbf{x}_l) + c \nabla \times \mathbf{m}_l w(\mathbf{x} - \mathbf{x}_l) \\ &\quad + \nabla \times [\mathbf{p}_l \times \mathbf{v}_l w(\mathbf{x} - \mathbf{x}_l)] \\ &\quad - \nabla \times \nabla \cdot [\mathbf{q}_l \times \mathbf{v}_l w(\mathbf{x} - \mathbf{x}_l)] \\ &= \langle q_l \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle + (d/dt)[\langle \mathbf{p}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad - \nabla \cdot \langle \mathbf{q}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle] + c \nabla \times \langle \mathbf{m}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad + \nabla \times [\langle \mathbf{p}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad - \nabla \cdot \langle \mathbf{q}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle]. \end{aligned} \quad (13)$$

Finally, using the fact that the total current density is the current density of the conduction electrons plus the current density of the molecules, we obtain the average current density:

$$\begin{aligned} \langle \mathbf{j}(\mathbf{x}) \rangle &= \langle \sum_{j(ee)} q_j \mathbf{v}_j \delta(\mathbf{x} - \mathbf{x}_j) \rangle + \langle \sum_l \mathbf{j}_l(\mathbf{x}) \rangle \\ &= \mathbf{J}_F(\mathbf{x}) + (d/dt)[\mathbf{P}(\mathbf{x}) - \nabla \cdot \mathbf{Q}(\mathbf{x})] \\ &\quad + c \nabla \times \mathbf{M}(\mathbf{x}) + \nabla \times [\langle \sum_l \mathbf{p}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad - \nabla \cdot \langle \sum_l \mathbf{q}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle]. \end{aligned} \quad (14)$$

We now have expressions for both the average charge density and for the average current density, and we are in a position to derive the macroscopic Maxwell equations.

VI. MACROSCOPIC MAXWELL EQUATIONS

We make the following identifications⁹:

$$\mathbf{D} = \mathbf{E} + 4\pi(\mathbf{P} - \nabla \cdot \mathbf{Q}), \quad (15a)$$

$$\begin{aligned} \mathbf{H} &= \mathbf{B} - 4\pi\mathbf{M} - \langle (4\pi/c) \sum_l \mathbf{p}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle \\ &\quad + \nabla \cdot \langle (4\pi/c) \sum_l \mathbf{q}_l \times \mathbf{v}_l \delta(\mathbf{x} - \mathbf{x}_l) \rangle. \end{aligned} \quad (15b)$$

If we substitute Eq. (10) into Eq. (6c) and Eq. (14) into Eq. (6d) and use the above definitions of $\mathbf{D}$ and $\mathbf{H}$, we obtain the macroscopic Maxwell equations in standard form:

$$\nabla \times \mathbf{E} + c^{-1}(d/dt)\mathbf{B} = 0, \quad (16a)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (16b)$$

$$\nabla \cdot \mathbf{D} = 4\pi\rho_F, \quad (16c)$$

$$\nabla \times \mathbf{H} = (4\pi/c)\mathbf{J}_F + c^{-1}(d/dt)\mathbf{D}. \quad (16d)$$

It is well understood that since $(\mathbf{E}, \mathbf{B})$ are the averages of the microscopic fields $(\mathbf{e}, \mathbf{b})$, $(\mathbf{E}, \mathbf{B})$ are of more fundamental significance than $(\mathbf{D}, \mathbf{H})$. One should regard $(\mathbf{D}, \mathbf{H})$ as a shorthand way of

writing the expressions on the right hand side of Eqs. (15).

It is straightforward to show that the average charge density and the average current density obey the continuity equation:

$$(d/dt)\langle\rho(\mathbf{x})\rangle+\nabla\cdot\langle\mathbf{j}(\mathbf{x})\rangle=0.$$

This demonstrates that the approximations made in determining the averages are internally consistent.

VII. DISCUSSION

Our multipole series arises because of the existence of a small expansion parameter: the ratio of the size of a molecule to the size of the averaging volume. The expression for the average current density [Eq. (14)] may often be simplified because of the existence of further small parameters, for example, the ratio of the velocities of molecules to the velocities of electrons. The last two terms of Eq. (14) are proportional to the (usually) small molecular velocities and may generally be dropped with negligible error [the other terms of Eq. (14) also contain pieces which are proportional to molecular velocities and which also may be neglected].

However, these two terms can be significant if the system is moving as a whole. To keep the equations simple, we will give each molecule the same velocity, $\mathbf{v}_i=\mathbf{v}$. These terms will then give a contribution of

$$\nabla\times(\mathbf{P}\times\mathbf{v})-\nabla\times\nabla\cdot(\mathbf{Q}\times\mathbf{v}),$$

which is usually called the moving media contribution to the average current density.¹⁰ However, for stationary systems, the last two terms of Eq. (14) may be safely ignored.

Another term which is often not needed is the quadrupole contribution to the average charge and current densities. This is because $\nabla\cdot\mathbf{Q}$ is generally much smaller than $\mathbf{P}$. Therefore, with the above approximations in mind, the macroscopic Maxwell equations for stationary media are given by Eq. (16) with $\mathbf{D}=\mathbf{E}+4\pi\mathbf{P}$ and $\mathbf{H}=\mathbf{B}-4\pi\mathbf{M}$, which is the form presented in most texts.

Our definition of magnetization $\mathbf{M}$ contains contributions only from the orbital magnetic moments of the molecules and thus, has two significant omissions. First, the particles have intrinsic magnetic moments, and this contribution should be added to that of the orbital mag-

netic moments. However, this is a purely quantum mechanical effect, which our classical derivation cannot incorporate. We merely remind the reader that this effect should be included. Second, the motions of the conduction electrons may give rise to a magnetization (electron diamagnetism in metals, for example). However, the orbits of such conduction electrons will be quite large compared to atomic dimensions and must therefore be analyzed in a manner different than that used here. We will just be content to say that it is sometimes possible to write the average electron current density as

$$\langle\sum_{j(\text{ce})}q_j\mathbf{v}_j\delta(\mathbf{x}-\mathbf{x}_j)\rangle=c\nabla\times\mathbf{M}_{\text{ce}}+\text{other terms}.$$

The magnetization of the conduction electrons $\mathbf{M}_{\text{ce}}$, should be grouped together with the molecular contribution to form the total magnetization. The "other terms" in the electron current density may be due, for example, to a drift current which is present along with the orbital currents.

Finally, we must return to the problem of time averaging and ensemble averaging. We have stated in the introduction that it was desirable to obtain a general derivation of the macroscopic Maxwell equations from the microscopic equations using a minimum number of assumptions, and we have argued that this means we must use only spatial averaging. Now that we have obtained the macroscopic equations through spatial averaging, we may wish (depending on the physical situation) to time average or ensemble average these equations. This additional averaging will present no *formal* problem since time averaging and ensemble averaging both commute with time and spatial derivatives. For example, the ensemble average (or time average) of the equation

$$\nabla\times\mathbf{H}=(4\pi/c)\mathbf{J}_F+c^{-1}(d/dt)\mathbf{D}$$

will have the identical form, with the quantities appearing in this equation now being the spatial *and* the ensemble average (or time average) of the relevant microscopic quantities.

These possible additional averages are of course difficult to execute in practice since they involve the interactions within the system. Spatial averaging takes the configuration of the system as given, and requires no knowledge of the probability of the configuration or how the configuration will evolve in time. Although it has not been the purpose here to discuss the dynamical as-

pects of the Maxwell equations, it must be stressed that the averages which we have derived represent just a first step in the analysis of the macroscopic equations and that these averages become useful only when combined with an understanding of the interactions between the particles and the fields.

VIII. COMPARISON WITH OTHER DERIVATIONS

The analysis of Van Vleck¹ is similar to those found in most of the standard textbooks. Starting with the microscopic charges and currents, the scalar and vector potentials are calculated. A multipole expansion in the internal coordinates of the molecules is made, from which the average charge and current densities are deduced. This approach has two weaknesses. First, if the point at which the potentials are evaluated is very close to some of the molecules, the multipole expansion which is used for *these* molecules is invalid. The analysis is certainly correct for points outside of material media, but it is not clear if this method can be extended to points inside. Second, a method which starts with microscopic charges and currents, deduces potentials, and finally obtains average charge and current densities is rather indirect and might give the impression that the average densities are mathematical fictions devoid of physical import. It is clearly preferable to average the microscopic densities directly.

Our method of averaging conforms quite closely in spirit to the ideas set forth by Rosenfeld,² who in turn has presented a more refined version of the method used by Lorentz. Rosenfeld averages the microscopic densities over sharply defined spatial regions, a procedure which we have criticized and modified. Our modifications not only result in smoother macroscopic quantities but allow us to obtain the averages in a more simple and direct manner.

Mazur and Nijboer³ obtain the Maxwell equations through a procedure of averaging by the methods of classical statistical mechanics. The formal appearance of their equations is identical to those derived here. Their method has two drawbacks. First, it is rather strongly tied to the ideas of *classical* statistical mechanics and does not lend itself easily to a generalization to quantum mechanics. Second, as we have previously stressed, spatial averaging is often unavoidable (notably in

solids), and Mazur and Nijboer's analysis is not designed to cope with this situation.

De Groot⁴ also obtains equations which are formally identical to ours. His procedure is to make a Taylor series expansion in internal molecular coordinates of delta functions. The justification for this expansion is not clearly given. However, the expansion of a delta function produces the same formal results as our expansion of w , so that we might take the view that De Groot is actually performing a spatial average.

We believe that the derivation presented here incorporates the essential physical picture which has been advocated in the works of Lorentz and Rosenfeld, and at the same time makes use of the systematic mathematical analysis advocated by Mazur and Nijboer and De Groot. Moreover, our procedure may be extended to include quantum mechanical systems, and the possibilities of further time or ensemble averaging.

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¹ J. H. Van Vleck, *The Theory of Electric and Magnetic Susceptibilities* (Oxford U. P., Oxford, Eng., 1932), pp. 1-13.

² L. Rosenfeld, *Theory of Electrons* (Dover, New York, 1965), pp. 13-27.

³ (a) P. Mazur and B. R. A. Nijboer, *Physica* **19**, 971 (1953). (b) P. Mazur, *Advan. Chem. Phys.* **1**, 309 (1958).

⁴ (a) S. R. De Groot, *Physica* **31**, 953 (1965) or (b) S. R. De Groot, *The Maxwell Equations, Studies in Statistical Mechanics* (North-Holland, Amsterdam, 1969), Vol. IV.

⁵ G. Russakoff, Ph.D. thesis, Cornell University, Ithaca, N. Y., unpublished. For quantum mechanics, Eq. (2) is generalized to

$$\langle A(\mathbf{x}) \rangle = \int d^3s w(\mathbf{s}) \langle \psi | A(\mathbf{x}-\mathbf{s}) | \psi \rangle.$$

⁶ For example, see the discussion by E. M. Purcell, *Electricity and Magnetism* (McGraw-Hill, New York, 1965), Vol. 2, pp. 344-347.

⁷ It is often the case that the center of mass velocities of the molecules result in a negligible contribution to the average current density. For such a situation, the following analysis may be simplified if all of the $\mathbf{v}_i$ are set equal to zero from the start. We will discuss this point further in Sec. VII.

⁸ A useful identity, $\nabla \times \nabla \cdot (\mathbf{q} \times \mathbf{v} w) = \nabla \cdot \nabla \cdot (\mathbf{v} \mathbf{q} w - \mathbf{q} \mathbf{v} w)$.

⁹ Our results are the same as those of Ref. (3). For a slightly different set of definitions see Ref. (4).

¹⁰ For a more complete discussion of these terms, see Ref. (3) or 4(b).